ActInf GuestStream 052.1 ~ A ElSaid, T Desell, A Ororbia "Ant-Based Neural Topology Search"

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hello and welcome this is active guest room 52.1 on August 10 2023 we have Ahmed al-said Alexandra aurobia and Travis de cell is going to give a presentation and following we'll have some Reflections and discussions so thank you all for joining and amen to you for the presentation okay thanks so much for having me um today I'm going to this to discuss um our um the methods that we came up to solve new architectural search and new Evolution problems um the methods that are um and Colony optimization based Solutions um so as the title ish is mentioning um and Colony optimization for neural architectural search and Revolution um this work this work is um collaboration between me uh my uh previous advisor dried the cell and my code advisor um Dr Ora and senator I'm currently um an assistant professor in University and um North Carolina University North Carolina Wellington um and moving on to the next slide um so as an overview the things that I'm going to discuss today I'll try to take um give um bird ice view for what is the revolution um why we why why we need it and from there I'll just try to discuss um our methods that is and that is based on and calling optimization which is called end-based neural policy search or ends for short and after that I will discuss how we Advanced with this idea um by um introducing a continuous ants or cancer control which is um which good thread of down this research space and replaced it with A continuous space and after that I'm going to also say what we did with the three-dimension The Continuous ants to have a back propagation free class and later I will um discuss three of the points that we are answering for um future work so in the machine learning uh learn as the structure the neural network structures get got deeper and deeper um people were trying to optimize the structures to have better performance and people and people in different Realms or different um problem domain um used to um borrow the best performing neural architectures or structures and try to modify a little bit uh to work for for their problem and they tried to be to tweak some of the pictures of the of the structure and compare um these different tweaks and then they say that we found the best performing structure but to actually find an absolute um open structure it it would uh concerning that solution it would be a NPR problem because to reach that solution we they had to um to to try all the combinations of the different structural elements right and this is and because we have massive um structures in these deep neural networks um they could be an NPR problem because we don't have computational uh Power or enough time to actually um train and test all of these structures all these structures constructed from these different combinations of construction hands so the alternative way to do this is to actually try to do to apply a meteoristic method and to convert to an ear to open solution that is much better than not optimizing the structure or relying on a kind of like a random search right a random search can get us um better performance performing neural network but it's not going to give us um an a mute to Optimal solution or to converge through near to Optimum solution so I mean heuristic method would give us an automated method um and and also we can converge to new document structure near to open solution to this uh structural problem um so the the way that people approached Nas was by trying to mimic um how optimization is done in nature right so the first method the the first way the thoughts are uh it was um to trying to mimic how um living organisms um evolve in nature um using um genetic based algorithm like the Darwin kinetic evolution and it started with a neat um or um it's it's a short for um uh new Revolution uh foreign that concept was also applied for in most of the nas on normal motion methods and exam is one of them tries to uh came up with this method and it became one of the state of the Arts uh instead of the art methods in uh in in Nas um so in in in in such methods we tried to again mimic genetic evolution by introducing new structural elements or removing structural elements or altering the structure through the evolution process through the evolution iterations and through the evolution Generations so we can apply mutations by splitting edges or adding edges or display disabling edges and this we display Edge we disable edges will disable some structural elements so that we don't lose that component so that we can later on use it kind of like um a dormant uh Gene in a genome right so that we it can appear in later generations not to get rid of totally so we can in mutations I can disable edges we can enable them later in later generations if we found that we want to try um this option um we can also add recurrent edges or remove recall or or or enable or disable the current edges uh we can we can split nodes we can take some of the nodes that in in the previous generation of um uh and we can just uh split it to two nodes and then take the edges uh connected to that node and um try to divide it between the new stat was generated from the previous node in the previous uh generation uh also we can add nodes from to the structures and the structure so all of these are part of the mutation classes in the genetic process um you can also disable a note if you don't if you want to try to just get rid of one of the notes and disabling a node or multiple nodes will also disable the edges connected to that note um beside mutations we are the other side of um a genetic process is to do crossovers where we have two of the best performance uh population uh made together to bring an offspring and Offspring will have um collection of the characteristics uh of coming from their parents um so it will take some of the characteristics from that parents and other characteristics from the other parent hoping that this will give us a bad performing neural network uh or platforming generation um so the main problem sorry so the main problem with with um uh genetic based algorithms is that they start with minimal structure like we can see there meaning that inputs and outputs and um you starting from the the optimization with this minimal search space can trap the method in a local Minima through the optimization process so we were thinking how to get rid of this obstacle and by having a bigger or larger space to start with and then we can sample some solutions from that larger space and we were looking around and we we we concerned end calling optimization and and I would say why that I'll try to introduce um the method first so the method was first introduced as um graphic optimization method um graph optimization method sorry um it was introduced in mid 90s by Margaret River applied this method um uh in on a terrible sales Mass problem and the problem is mainly about a travel salesman who wants to visit number of cities in the country do you think it's George's staff and considering this problem if we have different number and if the number of cities grows then that combinate the implementations of these numbers are updated that we have to consider to find the open solution um if this number says gross then we will end up having an npmr problem because we did we won't have enough time or or computational power to have this um this compact this explosives solution um done or this pixel step search done so he here from his observations um to advance in nature he found that he can apply how they Forge to find food in nature and um and uh and and then take this this concept that this observation applies in an algorithm to find the not the optimum path that leads from uh from one point and to visit all the cities and in a short shortest plan um so the methods this observation um so this line in the comic slides will just try to give us a picture of how hence um pourish in nature to plant food and then how Market reggae took that concept to apply it and uh in that Charles's hitman's problem so um hence observance funds that and go out from from their natural plant food um the the in Detroit different directions right um and when they find food eventually find food they will take some of that food and then they will go back to their nest and in the way back they will deposit some other substance called a pheromone um that the so they deposit that substance so that they communicate that through the foods resource with others so actually other ants do exploit this for more other substance and when they sense it they try they will they follow the path that the end that the first end took from that uh from the police was to the nest um hoping that they will find food at the end of that time and and when they actually find food at the end of the the path they will take some of the food and do the same thing they will deposit some more from lawn on the same path they can get more appealing to other ants who take it so that they can bring more food to the nest so the this in this process shows us the exploitation uh behavior of ants but again from sometime from time to time other ants also try to explore some other uh full resources potential food resources and they they kind of resist uh following the the formal uh phrases uh and and they try to find to go away and find some um new food resources for exploiters they also explorers and and these two concepts were um used by market the rager to kind of like right balance the search bit for the better uh the better or faster path between um the the the the the the the the the cities for the Trump salesman's problem um and the third thing that they observed also uh in Helen's Sports and in nature is that the odors for more the other substance the hormone also evaporates so whenever a path to a full resource is not appealing anymore with the full resources are exclusive no more ants will take that path or when they take it and reached it and the the the full resources that is exposed that they will not take the same path to the NASDAQ and they will try to wonder and to find more new food resources and because they not go in that path again and not depending any form on pheromone on that path the hormone will eventually evaporate and disappears and and making it less and less uh appealing for other ants to take [Music] so that's what Marco Durango was looking at when he thought of the travel spaceman's problem um he wanted he applied that uh for the federal salesman's Problem by making one agent um try this different tasks and then uh comparing in it through each iteration so that agent will take a path between the the cities right and then it will compare it the the the the length of that path to the previous uh experience with other paths and if it's shorter it will try to deposit for months on the segments of that path and eventually he was hoping and he was right about when he was hoping he was hoping that it eventually the the more um that the total pattern the shorter shorter segments that give the ultimate shorter um path um had more info more and more from one deposits making it more more appealing for the agent to take it through the iterations so we we thought about this concept um and we thought that it's very appealing to apply it for a an Nas problem because solution was applied for uh graph optimization problem and um neural networks are in their sense um Direction graphs so we we also considered ankle optimization because it's full torrent decentralized and scalable and it's also easily traceable and and and think going back to being decentralized uh it gave us it make it it made it it made this uh um method a perfect candidate for a pal and high performance Computing uh um and so which will eventually accelerate the optimization problem and I will in the next I think in the next Slide the slide after the next one I'll discuss how we um exploit or use this characteristic of end connectivization um to accelerate the solution uh that we came up with or the method that we came up with which is ants and games so um this scheme or the scheme of ants applying ants or end call accommodation uh in a neural or sorry in neural neural architecture search is depicted depicted in or Illustrated in this um uh flow chart so we start off with a massive search space expressed in superstructure which expresses or um it embodies a neural network that is so massively uh connected meaning that each node in that superstructure is connected with the other nodes through edges and recurrent edges backwards and forward and backward current edges and then we let a number of Agents swarm over the structure uh from an input node to an output node so each one of these agents will pick an input node and then it wanders from that node uh through the connections the current edges and recurrent edges and between the nodes in between hidden layers till it gets and picks one of the output nodes and then we take all these Paths of the different agents and we put them together to format our a structural and a neural network structure and we take that structure and drain it and test it and then compare its performance to the population of best performing in neural networks through this performance structures and it's better than the worst in the population then we reward um the path that the ants took the agents took over uh in the superstructure we worked with hormone um and so that it may make these paths appealing for later iterations through through the evolution process or the or the optimization process that's if the the durated structure is best than the worst in the population if not if it actually was worse and then the worst in the population then we discard that structure or that in your network and we don't reward any of the path that ants took and also the thermal evaporation will help us get rid of the the pheromones that were deposited on the edges that are not giving us better and better structures and again because the end colony is decentralized we exploited this by having um an asynchronous solution or asynchronous Evolution um we had a main process that took care of um generating the new structures and also updating the the population the test performing structures and updating the poor moon on the superstructure so the main process will generate generate um structures and send them to worker processes the worker processes will train and test the neural network on the data that we have for the problem and then send the results back or the thickness of the neural network to uh the main process and based on that Fitness that being process will um either discard it or um we'll take that this Fitness and compare it to the best performing in the population if it's better than the worst it will reward the path that ends took on the superstructure by depositing motion mode or if it's worse than the worst in the population will just discard it and it will keep generating um new structures and sending them to processes because the the training uh which relies on that propagation uh is the most computationally expensive part in this process if we have a number of worker processities that can work in parallel uh to train and evaluate these uh neural networks this newer new structures we can speed up the process right by training and evaluating different structures at the same time internal in in asynchronous pollution scheme um this is actually um this is an animation but it's not working in this version of the slides um because we're using a PDF but it's mainly a structure where um you'll see um edges or connections between these nodes um fading and or having darker colors based on the per month values through the iterations so each frame in this um uh in this animation is kind of an update for the pheromone value of the the edges based on the performance of the version of the of the neural network that were generated by the Asians when they swarm from the start node taking one of the input nodes in the middle layer and then from there going to one of the Hidden layers in this one hidden layer that we have here and from there going to the output so um that was the concept of applying an ATO and current optimization and um Nas and this now I'm going to talk about the actual method that we came up with so it's more dramatic and more powerful um neural detection search methods uh more comprehensive if I can say if I may say that we we we opted to apply the method and uh on um we require neural networks because they tend to be potentially larger than um other C other or neural networks structures because of their current connections uh so we thought that if we started this problem build the method or the constant applies to any neural network but applying it to recurrential um made it more appealing challenge to for for measuring the performance of the method that we thought of so this line in the coming side will discuss the different heuristics of the methods of ants the characteristic is superstructured itself um and as we as I mentioned before it and it's a method search space and at method as possible to be handled with the the the machine or the hardware that we're working on um the superstructure consists of a neural network that is massively connected meaning that every node in that structure is connected to the other nodes they are or through edges forward edges and recurrent edges backward forward edges and backward Corner edges um this simple structure that we have here represents one of just the concept of the superstructure that we have we apply in in ants here we have three input nodes uh three hidden layers each have three nodes and one output node in the output layer and we are just showing one node connected to the other nodes through edges which are the ones represented in green forward recurrent um the the edges are the one in Gray and then forward current and backward current in the green end uh in the red and the concept of recurrent edges um might be a little bit confusing if we look at it in this example because how can an edge um appear current coming in and out from the same node or of the the notes in the same time step but this structure might make it more uh clear um so here we have um a structure that three is pretty much similar to the one we saw before but here we have three input nodes two hidden layers there are three in layers in chapter three nodes and then output they have two notes um and then we have also three time steps the current times that t zero and the the previous times that T minus one the one before that T minus two edges here are Illustrated using the the solid black lines um and and of course these edges are present in the current time step that is going to propagate through the neural network and then the current uh connections these ones are going to bring information um or provided information from the previous time steps the previous um uh inputs or previous uh data that was fired to the nodes in the previous language and these are current edges are detected here in red and orange the four ones are better in in red and orange the reds are coming from T minus one and the orange are coming from the T minus two and the backward current edges are the dot lines in blue and green and um and you can see that they are going backwards so they cut the go backward and to backward layers but the uh but we but because they already current we can do that because they are they process information they bring information that already processed so we don't have to worry about improgating information back through time or back through the structure but we can do that if it's coming from previous timestamp the second heuristic and ants is that Colony weight sharing we wanted to use um then so instead of initial branding initialize the weights or the synaptic weights and generated neural networks uh we wanted to use the the train weights to initialize the weights of the newly generated uh neural neural networks we did that by Saving these neural networks on the superstructure uh we used the last um equation here to do this update so we balance between the ways that were saved previously and the ways that we were uh are coming from the trained or evaluated neural networks and we also used two Chinese to do this update we used um a fixed parameter file or if we let find that that was the first strategy the second strategy was to get Fly by applying these two equations which relies on the performance of the neural network that was previously generally oh sorry the the previously related entrance so it relies on the performance of the neural network that was trained and validated or tested so if um if it if the promise is was good then we will let this uh the weights of that um uh neural network to contribute to the the to contribute more to the uh initialization of the way so the newly generated RNs if it's not performing that well then why will this equation will not allow it to contribute that much okay the third meta heuristic is multiple memory cells so um each at each node um uh when when when when an agent or an ant reached to a node in in the superstructure it will do um a local search um to pick uh the type of the new in the neuron or the type of the node uh from these three different types of memory cells so in the generated RNN um the generation structure the nodes in the structure is not all the same they will be different based on the local search that that the agent will do or the end will do at each node they reach through their path from the input to an output in the superstructure yeah the fourth majoristic is the multiple anti species so we applied um different species uh or came up with different species for the ants um the first specie was the ones that will Traverse over the edges only the edges so they would go only forward through the edge at the the edges of uh the neural network of the mass the and these ones are going to define the number of nodes in the generated structure I will also Define the types of the nodes uh in the superstructure and when they're done with their work then the social or the second species is the social ants well Traverse between these nodes but they will use the uh their current edges to do the to do to move between these nodes so they will create them the recurrent edges for them really generated RNA and we have two different species for these social ants or three two different subtypes of species one is um the forward so for so the forward uh social ends we these ones progress only over the uh that poor connections they go from the input to the output but over only over the the for the current edges and then the back where it kind of just all the backwards social ants these ones go go from the output to the input and the Traverse over the Recon connections and the way the reason we thought about these different species is that we wanted to control the tendency of the ants to wander in uh around in the superstructure exploiting the the the the convoluted um mesh of um recurrent connections um so we wanted to control that so we came up with this strategy so that we can just Define the structure using the explore ends and then the recurrent connections can be defined after that using the social answer the fifth the fifth um meteoristic is the regularization of uh promo statement again we want it to um give advanced in an incentive to bring to bring sparser and also well performing neural um we by just penalizing them if they bring if they constructed um denser or bigger structures so we added this regularization term and um formula that updated the hormone value uh and as you see the regularization term relies on the performance the Ada here is the performance of the neural network and also relies on the size of the structure um the last the thick and last one is um is is is jumping and jumping ends which is we want to experiment with if we let the ends of um if we let the names um jump over layers when they move through the uh through the the superstructure um if we let them jump over these layers to construct the neural networks compared to if we restrict their movement uh to jump one layer at a time how would this uh end up um performance wise if they will give us parser and while performing uh structures or they will or this jumping would hurt down the performance um by keeping us weaker uh structures we use the time series data uh that can belong to coal-fired power plant uh we divided the data to have uh 7200 um records for training and testing and here the plot shows that the data and we can see that it's non-linear and um and it's a cycle and um and not seasonal so it's hard it's a hard problem for a non-um uh neural network solution or a regression linear regression solution so the input consists of 12 parameters when we when we we were trying to predict only one parameter the flame intensity experiments covered all of the heuristics of uh of ants um giving different values for these different parameters um the superstructure is consist of 12 really good nodes three hidden layers and chapter 12 notes and one one output node in the output player um the recurrent connections spent over um three time steps one two and three times steps um until the the superstructure had 49 nodes uh 9924 edges and almost 3.5 um recurrent address so if you unroll the structure over 30 72 time steps uh in in that propagation through time will have about 3.5 sorry 352 um thousand you know which about 6.5 million edges and 20 about 26 million uh recurring edges we in the experience we also compared performance of ants uh using the same data set uh we compare it to exam and need so exam is the state of the art in um neural texture search which is um genetic based um methods we also compare it to neat because it's like a benchmark in the new Revolution and Nas uh well we and also we convert it to fixed structures unoptimized structures that were one two and three uh that had one and two and three hidden layers and also different types of memory based cells the experiments covered 1600 experiments to cover all the combinations of premature heuristics and the heuristics of ants um each one of these experiments was repeated 10 times for uh uh for statistical analysis Illustrated well 2000 rnns for each experiment each trained for 10 Apex in in total and it's generated about a dream related training and evaluating 32 million uh rnns it took a month and well and a thousand CPUs to finish the experiments the results that we got showed that ends up from of course out from the unstructured unstructured structures uh unstructured uh sorry unoptimized an optimized structures and it also outperformed uh need and then some of the combinations of ants outperforming sense so exam here is the fourth from the left and we can see that the the mean absolute error or some of the versions or the combinations in health and heuristics uh out from the exam so we we try to look to do some statistical study or the results we got from ants so we tried to look at the top uh performing neural networks coming from in our results so we try to look at the top 10 25 120 200 250 and 500 results and we could look at um the contribution of these heuristics in this best performing uh neural networks or structures we found that these heuristics contributed effectively uh in most of these results but the more the the thing that was really um intriguing for us or or surprise surprise us is that um the that we saw that that their current connections um disappeared in this results because the best performing neural networks um didn't have that much of Recon connections um we would that mean meant for us that the the the memory based cells did the job for recurrent information coming from the previous times test but we wanted to just to expand on this layer but so it's on our list on discussing on on our future investigations uh so this is just a summary for um the achievements of ants based on the results that we got from our experiments foreign exploiting recurrent connection prove successful because um the the regularization component of the organization heuristics from uh it may give us better results showing here in this table and also the jumping ants jumping ants here gave us better performance compared to non-trumping ions and uh and the realization also um also the the the the data which ring strategy also proved um effective when we if we look at it there is also here compared to if we don't apply uh wage hearing okay so ultimizing and strategies uh are generic so the the strategies that we use are generic enough to apply it for any um any problem or solution that is end Colony based and end call optimization based the Vermont deposition uh that the method that we came up with is also the value it wasn't introduced in any illustrator previously creature um and the performance of ads compared to the other uh Benchmark and state of Art in the realm is also um remarkable going forward we thought that and was did gave us a good resource but the main drawback of ants was that the the the creature space um so ants worked on this massively connected uh uhmetically connected superstructure but it's massive yes but it's still uh discrete um ants can move freely between these um nodes and or they are forced to move over between these notes over these uh predefined uh connections whether they are forged edges or the current edges um we thought of removing that continuous search uh the streets of space and replacing it with A continuous space so we designed a 3D search space uh where um the the search bit had like layers representing the length the time lengths um and then ants can jump over between these these layers to have to give us the the recurrent connections and on each of these um layers hence will just give us the forward and the port sorry the edges between the nodes and the edges between the nodes um so in this line in the comic slides we'll show an example of how I move in cans or continuous and um or continuous ends um so the an agent or an ant will just start by picking up one of the layers that will move on this is done in a discrete fashion and once this is done it will then decide if it's going to do an exploitation exploration movement and and this um example we decided to do an exploration movement so it will um decide the angle and the radius of its next allocation on on that layer once that's done it's going to go forward and to that location and then decides if it's going to do an exploitation or if the next move will be exploitational exploration move and this example exploitation move so it will try to exploit the formal traces the Vermont that was previously deposited by other fans on in the search space so it will use that sensing radius that's something that was is previously defined through each end and then it will find the center of mass of the hormone traces and then once it finds that it will calculates that Center of mass of the hormone it will consider it as in its next location and then it will move to that spot and then it will decide about this if its next move will be an exploitation exploration move an eight and and at each location before it's deciding if about the type of the step if it's exploitational circulation it will also decide that if it's going to stay at the same level at the same time like or jump on uh to a next timeline um if if the jump or the movement is on the same level um if the if the movement is on the same level the end when if the S is doing an exploitation movement it will only consider that from one phrases that is uh um ahead of it um because it can't move backwards on the same time like otherwise it will be doing backwards step which is not allowed in in your Networks but if it's going to um another time lag um above above layer then it can be it then it's going to do a recurrent Edge and and so current edges can go back in time oh sorry back in the structure in the previous slide uh layer so now the ant can consider all the formal traces when it's tense and radius the ones that are ahead of it and the ones that are behind it so in this example he is going to consider um in this step is going to consider all the remote traces when it's sensing radius could calculate the center of mass and then consider it as its next position and then to keep doing do you keep doing this till it reaches to the proximity of the output nodes and once this is done it will consider it will decide which time which output node it will consider as its final position in its path from an input to the output um then other ends will do the same and then we'll have different paths from some inputs and some outputs and then accounts will take these paths and then try to convince condense um the nodes so that we don't have so much nodes and and that are very close to each other so the one nodes that are within certain proximity will be uh clustered together using DB scan um to have less number of nodes and then the path will be the paths will be taken to collected and put in a structure a new enough structure and um then sent to a worker process to train and test and then compare it to the population the best performing RNs and the plus will be almost the same from this point will be almost the same as ants once the structure is constructed and the training and testing process will be the same as the ones that we discussed in dance um so this is this was also an animation which show which shows the the how these paths are taken by the ants uh look uh from an input to an output in the 3D it's 3D search space so as if we look at ends uh the ants have uh only eight uh tunable hyper parameters compared to When comparing this to ends and exam it's half of the number of hyperplanners and other methods these hyper parameters are the number of ways layers of the search space the number of kind of legs how much how many of them we have in search space number of Agents number of ants which is similar to what we similar we have a similar uh hyper parameter and ants the the sensing radius of the agents or the ants um and the agent's probability to create a new node uh which uh represents the exploration instead of exploitation of the parameters or all that from one uh back traces in the industries um node compensation um parameters or uh uh factors that the the the the the variables that are represented in the DB scan are considered also a hyper parameter hyper parameters and um in cancer um for one updating parameter and from one volatility primer and these are also present in um and and SEO based problems uh solution sorry um the experiments used three times years um different number of parameters and different sizes um the results showed that hence the results that we got from cans were very uh competing with um ants out and exam um they weren't necessarily um better but they competed they were um but it also uh did not didn't do very well uh on one of the database their data sets but it's uh comparing the that was comparing the performance of as the main absolute error but uh comparing cancer results uh from the uh from the size of the new networks that we got from from Cairns we saw that case our structures were um sparser it so the performance was competing with uh hence and exam but the structures were also were compared compared to ants were much uh sparser uh similar to Sam uh the the size of the structure that we're getting from exam remember exam is um genetic based meaning that the start the structure the struct optimization process using the minimal structure elements um but it was susceptible to it was susceptible to local Minimus trap traps um and tense is not suitable to this problem but it also gave us um smaller structures and with performances that we're competing with other message examinants so the the advantages of cancer is that it's it has an unbalancer space compared to ants um result results compared uh the results that came out were good compared to ants and exam hyper parameters the tunable hyper parameters are half of the of these in examine ants um and also indirectly encoded the neural topology to 3D search space so this is one of them maybe the contributions of important contributions depends um then so so far aims and cans are solutions um that applied neural gold architecture search meaning that they did not optimize the synaptic parameters of the neural networks during the optimization process or during the evolution process um we thought we thought of actually making ends capable of doing this as well so to train the neural network or optimize the synaptic weights uh the weights of the structure of the neural networks during the optimization process the structural optimization process so we added a fourth dimension to the search space which we embedded in uh embedded the weights of synaptic parameters um in in that map this terminals in that new dimension we also uh wanted at the ends to be self-aware and try to evolve themselves through them the the evolution process so that they can um adapt to the changes or like um adapt themselves so that they can give us better performance um the the we want them to change their behavior of their characteristics like the sensing radius uh they had before for example they can be a variable that can be can change through the iteration thing evolve through decorations um each end can change their uh these characteristics uh as they are as the evolution goes on progress based on the performance of the neural networks that they are generating so um the the advantage of doing this was that it eliminated the back propagation process which is the most compilation completely expensive part in the evolution process and in the Pro in the in the in the method that we use so far in the exam ends and cans um so eliminating that that propagation gave us much much faster Revolution process there is also got um I will discuss the graph on the right hand side first which discusses the thickness or damage meaningful error of the results we got from back propagation free cancer the four dimension cans compared to the the normal chance the back propagation cans and um ends um so the results showed us that and again on a particular database um that that propagation uh canes and that propagation free chance um did a quite a similar job um and but they were both better than ants on this particular database um but they actually main contribution or the main advantage of applying case it shows up in the graph on the left hand side because we can see that if you compare the results um based on the time of the evolution that based on the evolution time we see that um hence with much had they took much much less time than the back propagation version of cans and uh the ants using these different um and number of amps also this graph shows how fast and that propagation pre-cance compared to the normal cans um we could so in this figure the Curves in the data lines are shows the time um that back propagation free cans and cans took to prepare or generate the neural networks or we can see that that propagation free cans took more time uh to prepare or generate neural networks compared to Mac propagation cancer the normal case and that's because it has the fourth dimension also has to revolve the ends of the agents through the iteration so it needs more time to do that um but um the other curve the other two curves thing in dashed lines here are the lines that shows the the time the amount of time it took for the two methods to validate much um train and validate the neural so Mac propagation free cans doesn't have to train the neural network doesn't do um uh that propagation so it you can see that at the time it took is in an order of magnitude less than um the back propagation version of and that's these solid lines shows the cumulative time took for both methods and of course it shows that that propagation precances took much much less time than um the back propagation three cans took much less time then the back propagation cancels um the future directions that we're just in the future points that we're concerning for uh sorry the point that we're considering for future work are to turn in against to a complete continuous space so um as you saw um we we the the the third the search space for cans is not purely um uh continuous because the time next are represented by discrete layers uh we want to replace this with the continuous Dimension continuous um layer representing the lag uh the time lag in in rnns however this is a little bit challenging because the num the time length has to be uh known prior to uh any optimization process because um other than that we we we will be in mapping the whole uh we the whole time series uh as um time lapse and then picking the time next from them which is not um feasible so this is the first thing we the first point we're considering to investigate for in future work the second one is to um investigate this finding that we found we we have we found in in in in the results um in ants where the recurrent connections disappear from the best performing uh structures um and the theory that we have is that the memory based cells replace these connections to give us the information that we need from the past 9 steps um so that they were more effective uh more efficient in doing this compared to their connections so we have to do this get expand on this and investigated more um the last thing we want to consider also mean this is one of the top things that we want just these three points on the wall the top on our list um the third one is to actually um you consider one of the the concept that was coined in a book by Dr Deborah Gordon um which we where she mentioned that um the living um organism in in ant's world is not are not the age themselves but there are the colonies the colonies are are the organisms that are um the the the start and they grow and and they interact with the environment of the ecosystem they interact with them with other colonies and they die at some point and the so the ants themselves are not organized they are the cells of these organisms The Colony organisms so we want to take that concept and apply it in our methods to have a number of colonies living together in parallel evolving and communicating with each other so that we and we want to see if this will give us a better performance but because after all we're trying to mimic nature in our in this in in the solution we we we're trying to investigate um so with this um I'm done with my presentation and if you have any questions awesome wow what a great presentation how about you Travis and Alexander either of you which want to go first please feel free to give an introduction and any primary remarks that you have on okay I can say something I don't have too much to add I think I'll go around and did a pretty good job you know going over the core bits of the work um I'm Alex rorbia I'm an assistant professor in computer science uh in affiliate faculty affiliate professor in Psychology and affiliate faculty in computational neuroscience at the Rochester Institute of Technology and I work on a lot of stuff but rain primarily predictive coding active inference variational free energy a lot of the stuff that's actually of interest to this group and uh uh yeah and this was a particularly interesting project for me because a branch of my own research is working in neuro-evolutionary methods or even just nature inspired meta heuristic optimization and uh you know when I got the pleasure of working with Abdul Rahman when he was a PhD student at RIT you know we you know we talked a lot about uh the and colony-based optimization approaches and I encouraged him and to do to look into sort of like the origins and also I try to understand actually how physical ants behave so that was always fascinating to me um I don't have too much to edit in terms of the technical Parts I think he covered all the core results the only thing I I like to think about and I'm actually more fascinated to to hear from the active infants Institute their interest in the and Colony methods and particularly what was interesting to them because uh thinking about what does and calling optimization how do you view it from like an active inference perspective is I think particularly interesting and thinking and I even had a thought I don't know if this was a question among the audience's mine do the little ant agents or the can't agents that I'll build around and described earlier um you know what is there a way to start to view them as like a multi-agent system that's optimizing some variational free energy quantity uh and you know because you know a lot of even like Carl's work I mean I collaborate with him you know sort of touches into this area of like well what happens with collective intelligence and societal organization and you can kind of look look at free energy from these very very high level viewpoints all the way down to fine-grained Cellular activity and so I'm actually more curious to know when Daniel and anyone else in the active entrances who can sort of explain they're particularly interested in and based optimization and meta heuristic optimization I'm curious to know that but there could be some interesting viewpoints of you know like what is the the free energy I think the ants themselves are very simple I mean we have made them more intelligent I know of the Roman and I talked at length about well what if we even made them for example have like a reinforcement learning control system and you could even imagine well now what if the cans themselves uh you know engage in a form of active inference themselves what would that look like for the system and they are optimizing their own free energy those are just fun little thought experiments I we have obviously not worked on them uh at least Abdul Rahman was never exposed by me to that part of my world uh in you know and biomimetic intelligence so those are my comments I'm not sure if they're particularly helpful but they're very general uh and actually more curious to know from the active inference Institute uh their interest in it and you know where does that maybe perhaps intersect or is it just like oh we like you know interesting topics and intelligence and so yeah thank you Alexandra Travis you want to say hello and give any Reflections on The Talk we're all coming hope somebody can see me a little bit in White Jungle here I'm hi I'm Travis's cell I'm an associate professor at RIT I'm also also the graduate program director for a master's in data science so if any of you are interested in any of this or data science please you know shoot me an email um you know I thought this work was really very interesting in a lot of ways so a lot of Dollarama was working on it um I think one of the coolest Parts about it is a popular neuro Evolution algorithm um that came after me is called hyperneat and it transforms the um discrete search space of neural architecture search into a continuous one um and it's it's shown to be pretty a pretty powerful method well this this version of ant colony optimization the new one does the same thing that makes the search space continuous but what I found was really cool about it was that um as opposed to traditional ant colony optimization where you have a graph and you just send the amps along the edges of the graph and you can and you take the best paths and construct a graph after it um here the answer actually working like ants in the real world will move a continuous amount of space not just from point A to point B um and actually dropping down pheromones and it's more like a real simulation of how ants would move around in the real world um and we're getting better results from it than some of the old older methods so um I think that is that really made me be kind of happy to see that and thought it made it as a very very interesting work that's awesome um Ahmed want to add anything or I'm happy to give a thought on the ants and ask some questions from the live chat oh I'm good um trying to think covered all of the things that we I wanted to say um about uh um I think just one thing that Alex mentioned that we give higher intelligence to the agent is something that we discussed and I'm pretty much uh open to that I've started actually to think about it but they didn't Implement anything yet however the the last thing that I discussed in the future directions is something that I actually started working on but they didn't start the experiment patients yeah so hopefully we will see something out of awesome well there's a ton of ways to go and isn't that kind of one of the fundamental questions like in an interactive setting either pure agent stigmergy interaction or multi-agent but ultimately mediated through multiple stigma G's with like reading and writing and error correcting codes so in that communication setting um I felt like the work generalizes along multiple Dimensions that previously approaches to multi-aging just didn't have those kinds of flexibilities like the continuous time feature and and several other features and I guess with respect to the ants themselves um I did five Summers of field work with ants in the USA Southwest in Arizona and observed a lot of foraging activity fat problem or that context or setting is really a fun one and a pervasive one across any kind of living system anything that's going to be active and uh living so why did you pursue foraging type algorithms overall and does this class include the interaction based methods with direct agent contacts that Professor Gordon highlights in the ant encounters yeah I I wouldn't try to respond to this but I I just learned to act to mention that the British thought about this map actually Travis started this idea about using and quantumization in your version our new architecture search by applying this method in simpler neural networks uh Elma and Gordon neural networks your Cardinal networks and I did believe from there and started working on my pieces my my patient pieces so uh we thought about this idea I think I I mentioned a little bit about why we used uh and quantumization in previous slide but just repeating that for the audience to make sure that I the God we thought about this idea because um anytimization was applied for uh as a graph this uh a graph optimization solution and we thought that why not and why not your networks since they are in business uh neural networks they are directed so they are directed grants are directed perhaps because the the flow of information goes from one direction to another Direction so that direction but they also graph because they are nodes connected with address and our ultimate aim or goal is was to optimize the structure by removing and adding elements States so that it gives us awesome and and what one way that

Professor Gordon and others have have talked about that bi-directional uh learning relationship between the computer science and the math and the analytical formulations and then the the field work and the actual behavior is because the thousands of species of ants they're working amidst a huge range of ecologies with all these different patterns of regularities all these different resource distributions and foraging it's amazing how General it is yet it's also just one of the functionalities that need to occur in terms of these like even slower processes like allocation of tissue to faster processes even of like response to alarm pheromone so it's like this one class of algorithms clearly scales across from view ultimately one some of these foraging algorithms are law enforesters they don't leave Fairmont Trails so it's like even the idea of leaving a single positive pheromone or leaving then but also there's things models can do that ants can't do like the time travel pheromone lossless perception High dimensional uh signaling profiles that can't occur just with like finite amounts of molecules and um now they are just persisting on their path as active agents within one generation with a variational free energy at the behavioral scale and then across generations with that evolutionary layer and the relationships between the neural network implementation and the active inference model they're kind of like two ways of describing slash implementing um yeah I'd be curious to hear any of your thoughts on where you see active inference coming into play or how do you see old ultimately similarities and differences between neural network-based approaches and active inference-based approaches are they same complementary overlapping you guys want to stick this one I think this might be more of an Alex question to be honest well yeah I mean if you want to keep it on active inference but I think Travis you're going to need to tag in when you want to get into the very specifics of like the actual and Colony details because again I kind of see Dan Colony optimization from a more global point of view I would say so to be clear I mean this particular work that I'll be so that way of the ram is not also completely blindsided by your question Daniel even though the name is in the Institute itself uh this isn't an active inference work so you know again while there are obviously as you pointed out lots of interesting elements like for example the fact that the ants when they conduct their exploration along let's just think about like the recurring networks and they're figuring out what nodes and what as uh explain the superstructure right as they iterate across with the pheromone trails and figure out what nodes I want to recurrently connect be forward connect skip connect so on and so forth um you you would say well okay what these ants are doing is they're engaging in epistemic foraging which is a key Concept in active inference or idid so that way I'll build Ramen and Travis are not also left behind by jargon uh

Epi Summit forging in active inference it's a big General framework it's like a biological neurobiological process to RL and epistemic just refers to kind of like the uncertainty Travis that you and I work on and the ideas it's saying well okay I want to understand my world and the more that I explore my world right there will be things that surprise me less but if I encounter some information that is really weird when I build a generative World model or a predictive World model that's very surprising I should probably explore that um and so of course I'm condensing the concept down into sort of like the exploration part of the explore exploit trade-off and that I know you know but uh that characterizes reinforcement learning so that's just what we mean when we say epistemic or epistemic foraging and obviously of the ramen foraging you know can be likened to what the ants are doing right they're exploring their environment and so I guess with that in mind so that way everyone's sort of on the same page here uh to get to your question about the differences about how does this work versus like your typical neural based approaches to active inference in I would fall into that category of oh I build neural models biological process models uh those are very much focused you could say at the individual level at least the ones that I am aware of when you're building for example even a back propagation based uh partially observable Markov decision process in uh you know active inference that's like a single agent right you're trying to build this construct that is trying to you know balance the epistemic quantity with its instrumental term which by the way another jargon term for you of the ramen and Travis that's just like your reward signal or your prior preference or a prior distribution over goal States and so the agent these agents sort of like deal with that trade-off but at an individual agent level now again I'm sure there's an interpretations of these from other perspectives the ant-based approach even though I would not argue per se that this has at least an explicit form or a connection at least that Bob the ramen has made clear to active inference but the idea is that this is like a multi-agent approach to active inference and so the ants when they conduct their epistemic forging which arguably is a very very simple model each and each Anton of themselves is you know essentially a bunch of you know coefficients and some hard-coded rules uh because they're uh their job is essentially to work together with their pheromone trails to figure out oh what parts of the superstructure are useful so I'd say that that's different and it lends itself in some ways of course you can make the ants more complicated and lose the benefit I'm just about to say you could massively parallelize this and this is one of those key strengths that I think is natural in for example a lot of actually transpired optimization algorithms and and Colony optimization is one of them is you know Abdul ramen's been using hundreds of CPUs and you can put these ants on their own individual processor and the communication that's you know occurring across them as they exchange information is you know through the pheromone Trails there's an indirect mechanism it's not terribly complex to facilitate and I'm sure that there's even better ways to go about doing like asynchronous forms of communication and further further optimizes I Know Travis and Travis can add to this has done things on like citizen science and uh distributed computing through you know the you know volunteer volunteer Computing and how you can distribute this through a massive Global asynchronous Network so you can imagine adapting the uh ant form sorry the ant colony optimization approach to some like distributed massively distributed version of active inference where you essentially have to write down that the variational free energy and I'm I'm putting quotes around this because again there is no concrete term written in out the ramen's work I mean because we haven't at least we haven't viewed this from the active inference perspective directly each ant is optimizing its own variational free energy but then there's probably a global quantity that sort of is really as a function of those pheromone trails in the individual hand agents um and then of course with the exploitation term or the instrumental or the uh you know what is the reward signal to give an RL term that's sort of driven by the performance of the actual agent on the each candidate agent on the task right of the ramen you compute like mean squared error when you're doing time series predictions so in some sense we have built-in you know uh a reward function that we use and again for those in the active inference group here you know you can use the complete class theorem and look at you know the prior preferences saying oh well the reward is actually technically a prior preference right it's like a log probability so with that in mind you could squint at and Colony optimization and I guess the big benefit comes from that massive parallelization that you wouldn't actually very easily if at all get with you know our single agent neural based approaches um and that might be an interesting place to build on and I'll stop rambling at this point I'm not sure if that was helpful that that was awesome even earlier today Chris fields in the physics as information processing course was talking about the classical information inscribed on the blanket which is like the pheromone perception and deposition pheromone modification and and perception sense making an action which we can associate with like the nest mate cognitive system so then there could be as simple as a pass-through for the nest mate could be any arbitrary relationship described with a blanket simple Nest mate sophisticated Nest mate um like another level of Time series modeling whereas there's the environmental time series modeling and that's just in the ants and then the fourth dimension is like that Quantum rotation which goes from the lower dimensional classical stigmergic screen into the quantum informational space and so that's one of the discussions on going in octinf right now is about well previous approaches to connect Quantum formalisms to macro let's just say neural phenomena based it upon the plausibility of like a molecular electronic bubbling up whereas just with research from decision making and statistics and just multi-perspectival modeling and all the issues associated with the physicalities information transfer the finiteness of it the quantum formalism becomes useful just by itself with or without Reliance on some other electronic phenomena so it's just a lot of very interesting connections like having the degrees of freedom on the blanket which could be noiseless and four bits or it could be noisy with this really specific thing but in silico you get to play it from both sides and scale things up and down and do these meta heuristics on top of that arbitrary space could be really simple for learning or it could be however much and and then just like the ant colony algorithm is is ultimately Federated through embodiment that property makes it a really useful candidate for biomimetry so a lot of times when people think about collect Behavior they're thinking about like the flock of birds and the school of fish and those are of course Collective systems and Collective behavioral and all these kinds of like complex systems properties can be studied in that type of system but it is also neglecting it at least an analytical degree of Freedom with the stigma G so that really opens up that both the quantum the classical information or both Niche modification and behavioral uh and cognitive modeling so just to add on because I think and and then also what the node is could even be heterogeneous or unknown or fit in in different ways or fixed through design processes so just like the ant colonies are flexible enabling them to live in all kinds of places make all these kinds of nested decisions to interact with each other that that flexibility is just like the tip of the iceberg of what we could even just describe because there would always be real environments we hadn't yet tried with ant colony so we really would never know the full extent of like all the repertoire and the Dynamics of the ant system but then we can just abduct into new mathematical statistical distributional Frameworks pull back to different levels of the learning in The Meta learning process and just start there and then um almost ironically or or um maybe the opposite of that like it could be applied to ant colony video data or movement data or foraging activity itself but it kind of takes inspiration and develops in parallel or in conversation so it's not like balance what real ants can do or you could constrain it so that there are properties that real ants have like they can only interact with in this certain way or like there really are only this many pheromones do model comparison so it's it's a lot of degrees of freedom I feel like you all are opening up with the ant colony modeling and also one of the challenging pieces of multi-agent simulation is kind of like the open-endedness with the design space so then it's very hard for even creative ideas like sometimes to find the right compute resources that that obviously are still even needed for for what you discovered um with the analysis here an awesome question from Bert in the chat I just wanna I just want to clarify just so uh make sure that I got the right term and certainly Abdullah Ramen and Travis you might want to look it up is when you were saying blanket you were referring to a Markov blanket correct yes so the technical definition of Markov blanket is when you have a Bayesian graph where nodes are the variables and edges are relationships amongst these variables for any given node of Interest we'll just call it internal States so these are not features of the world it's not tagged on to some tissue of a real Nest mate in the world this is something that's tagged onto or like a perspective we could take on any node in a base graph and then all the nodes that insulate it in the co-parents are known as the blanket in the sense that it's like one layer insulator and there's a lot of more discussion on it and the philosophical implications and all these generalizations of that but broadly the Markov blanket is just the inbound dependencies which we associate with sense making and perception learning attention and then the outgoing dependencies for the agents which we associate with like action influence in some Downstream pointing way thank you I just wanted to clarify that because I don't think Abdul Rahman and Travis might be familiar just with that terminology it's very uh forstonian you know very active inferency kind of jargon so I wanted to make sure that they got that from the physics point of view thanks yeah yeah totally great point and um ask a question now from bird supper says very impressive solving generative models with more generative models sounds very promising what about replacing ants with convolutions I'd like a meta learning algorithm like having a neural networks that learns how to optimize your other neural networks yeah I mean this concept I think is is interviewed at some point I mean in machine regular but we we wanted to apply for nature based uh method that mimics like the the mother nature which is the uh you know uh nature is the most efficient optimization Revolution system um looking at the results that we could that are there in those nature implying nature based methods uh they were Superior to any other method um and the results we got also which just pointed out that pointed that out that we so good performers are coming out from these results from our results and previous results from other methods as well so so I hop in here a little a little bit too um in the case of neural architecture research there's kind of a couple classes of approaches all right one is constructive so kind of like where we build larger and larger networks and try and keep your network size as minimal to try and find your optimal solution other types of neural architecture search approaches use like a superstructure and this is kind of how the the earlier iterations of this were where you have a bound of your search space and you try and find the optimal Network within that bound so one is like trying to build things from the ground up and the other one's trying to like trim down a big Network to a small Network um ask your question about convolutions there's a type of there's been a fair bit of research lately in what's called graph based neural networks which can which can use convolutions over like a discrete Graph Search space and can potentially produce other graphs and I believe there's been some neural architect architecture search work using this but one of the main and I think cool things about the approach here which is different from those is that even if you have a graph based neural network and you have your search base um defined as some kind of Matrix where things are off and on depending on which nodes are connected to each other you have a fixed search space which may not be big enough or it might be not not be the correct search space for this neural architecture search problem where this method here is all continuous right so um within this continuous search space you really have it gives the algorithm a lot of freedom maybe too much freedom but um a really open-ended way of generating a wide variety of neural architectures which if you pre-constrain your algorithm to work within a fixed discrete superstructure you may not even find them because they're not even in they're not even a possibility so that's one of the reasons we didn't we didn't go that route but there there are graph based neural artistic architecture search algorithms out there where basically you take the architecture as a graph you um train a neural network to spit out another graph that it might think is better and those use convolutions sometimes that helps that's awesome so convolution sounds like yes um and one thought on that is yes the ants solve all these incredible patterns and and kind of do amazing things amidst informational and physical limitations like like we all do having the ants be able to just make trade-offs within a task space and then have a dial as modelers to make that task space kind of like touching the pheromone distribution or metacognitive ants and all or something emulating essentially that um and for example the octave inference forward-looking and like thinking through other Minds that there could be a kind of cognitive colony so then that enables in silico total thought experiment colonies and through data driven processes also kind of keep continuity with that model perhaps literally continuity with the model and then connected to empirical which is something that that is very hard for agent-based modeling which as you kind of Point into often sets certain fixed axes performs like a sweep looks at one mechanism doesn't look at all these possible mechanisms of like learning and intra and intergenerational and all these time effects so how do you see this being used in different research or application domains wow well I mean we were the main use case that we're thinking of was new revolutionary architecture search uh it's for to apply it for other domains other than your Arctic search we didn't figure out that yeah uh I think Alex can explain on that I can hop in a little bit too I mean so basically this this type of algorithm if you need to generate graphs and you don't necessarily have a fixed structure for that graph and when I say graph I mean like a computer science graph where you have nodes images um so pretty much anything involving graph construction I think these types of methods no no that works kind of under the hood are you know can be represented as graphs and usually are so I think we're using it for neural networks because it's really popular but um you know there's other other algorithms out there like you know the traditional uh traveling salesman um there's like routing problems all that any type of stuff where you might need to generate a graph in a smart way let's go do that yeah take another point though I think what's really cool here and I don't want to steal up around those Thunder but you know his last point on future direction is um while the a particular version of this ant colony optimization search is running to find an optimal neural network a colony has fixed parameters that it operates Within but if you think of the colony as an organism as opposed to the ants being an organism you can evolve colonies that optimize how it the the ants themselves act so you can have evolving colonies that in a smaller sense also evolve or optimize what they're doing within their prescripted uh parameters for the agents they're generating so you can have like a Meta Meta accuracy is really cool it's awesome I mean The evolutionary account of of a why question for amp Behavior today one part of that answer is like because colonies That Couldn't under that regularity or constraints survive we've had a long long time to wipe those off the table and so every biological system has to have that kind of multi-scale ordering um in 2021 we made the active infra ants paper which was modified from an epistemic foraging visual attention task about scanning around and then learning a cicade policy that had to do with epistemic foraging but not leaving the trace and then the main modification to bring that active inference epistemic visual foraging model into the active uh inference ant setting was to add a pheromone rule just like you described even though of course again that's not the only pheromone rule but that's just the most simple pheromone rule that we can generalize from as you definitely have and there are just many emerging ways of of modeling those multi-scale active inference models so composing across layers which we might associate more with the kind of laterality of things that happen through interactions and then also as Mike Levin shows with kind of the time Diamond systems that have a memory retention components of some shape cognitive shape and then a pro tension um awareness or agency or other attributes you can use to describe that and that that's a statistically amenable way to describe things and then there's a variety of implementations on a given statistical problem or like Federated compute architecture it might be the case that you're not running like the pure Matrix multiplications that are shown in like the early Matlab code of active inference different components of machine Learning Systems might be kind of composed together also ways uh kind of abiding by those patterns of communication but then there's a level of abstraction that that um we can still describe but it doesn't mean active inference is going to be like kind of causing it so that's what gives a lot of flexibility and it's really cool that through your backgrounds Alexander and work and and these kinds of collaborations that like the active inference perspective on multi-agent modeling with all these other views can at least come together comparatively and then that is gonna I think be quite an interesting Interchange to apply this entire tissue type or Colony type thinking up above and within the models just a lot of degrees of freedom like you said could be too much I mean and I think there's different ways to I think you depends on how you want to take the ant metaphor and you know again it's it's kind of interesting some of the questions or comments that you're making Daniel and some from the audience about you know uh how does this you know think about it from a cognitive point of view I mean I do work in cognitive architectures of course again kind of from the whole single agent you know or single entity and you know modeling a single brain in its different regions but I think if you take you know a nature inspired uh optimization approach like the ant metaphor that Abdel Ramen uh you know you know latched on to and that's sort of like the way in which he formalizes how or it takes a principle of how ants interact with their world interact with each other uh and then you know mathematically model those particular Concepts step by step I think if you bend the metaphor and say well okay could the ant colony metaphor apply to multi-human agent systems right or other entities does it does it does the ant necessarily can it be generalized Beyond you know the physical creature upon which uh abdulrahman based his initial metaphor and that's an interesting philosophical kind of take to it and then how do you apply that to let's say building a multi-agent cognitive system and then of course says Travis was discussing with you and you were mentioning metacognition and you know you could think of ant colonies of ant colonies but you could even replace the word ant and just say well we have you know clusters of you know intelligent agents or however whatever degree of modeling we're doing because again I do want to emphasize that at least they can'ts and ants agents that I have worked with in the context of the ramen you know they are not in each and of themselves even I would argue if nothing else a very extremely simplified generative model or a very very very simple control system there's no neural network under each one because then you'd have to simulate computationally each one of these ants within the framework so I think there's always that good practical machine learning kind of viewpoint of well you know there's always a how do you simulate that in you know obvious working with CPUs it's not like he has an army of gpus to replace them with convolutional networks again if you have the resources this would be awesome but uh you know expense and money is another constraint on this planet but um I think there's interesting views and interesting directions one could go by taking inspiration from the ant metaphor and you know the concept of pheromones and translate them to other you know other real world signals and how for example communication patterns among you know other animal entities or other human agents and I think that that opens up an interesting perspective and if you're constantly trying to connect it back to you know free energy minimization and trying to say well how are we balancing you know the terms that you can decompose it into an epistemic and an instrumental and how are these balancing out and how how are these physical processes that we specify uh you know balancing those terms that's just a very interesting place to be and you mentioned you know active inference versions of ants and that's fascinating and of itself uh last comment I have is again the degree of modeling and what you are modeling like if you're modeling a society or organization you know that's one way you could use you know the ant colony framework if you will or optimization or meta heuristic optimization Frameworks to you know then cast you know uh you know a system any type of complex multi-agent system as an active inference kind of uh you know engaging uh process or you could go really low level and think about you know cells in a body or you know or units that make up organs or organelles and trying to say well can we use this to model that level of granularity within like a human or an animal entity right and I think there's some fascinating questions about how does this metaphor manifest itself at different time scales and different degrees of you know perspective right about how you're modeling what are you looking at what's the picture that you want to emulate and of course there's always under the hood this practical consideration of well okay the computational expense they allocate and are you able to actually run that simulation long enough because I think all the ramen incorrect me if by if I'm wrong you mentioned like one of the experiments I think was for the bigger systems took a month right of course this was on CPUs uh you know that that can get pretty uh prohibitive if you want to go even bigger than that but again I think it just depends on you know what Hardware you have to simulate the Sun yeah yeah and also uh I mean using high performance Computing is not always feasible for for smaller um so if we try to model the brain of ants like small neural network um you're using a GPU might not be like a feasible solution and Travis is the high performance Computing specialist here it's expert value but like sending data to the GPU and getting it back it's very time consuming and resources can continue so it will worsen the the the the the time consumption rather than solving it because communicating between the the main memory and the GPU would it had like a an overhead so it has to be a big enough problem to actually utilize this GPU in in such Solutions yeah so um when you have like you know the super large language models or large models for computer vision they do a lot of just massive operations on tensors which are basically multi-dimensional matrices right and when you have really big wide tensors they you can parallelize the operation really nicely across the GPU a lot of this work is based on uh doing like a Time series forecasting time series specification on on sensor data I think stuff from like Power Systems so you know in the input to a large language model might be a thousand or more um you know word length of a word embedding which is actually not not huge but it only if you go up to computer vision model the input image may be a thousand five thousand pixels and that gives you like actually a million inputs um when you're working with sensor Systems off of aircraft power plants that type of thing you may have 50 to 100 inputs and when you're working with this type of Time series data you don't need a massive super wide neural network and then if you add in recurrency where you have to do back problem over time and over other things like this you actually can't really parallelize it nicely on a GPU so for us the CPU is actually more more efficient we tried Abdul Rahman wrote a bunch of code a long time ago to put this stuff on gpus and we found it was quite a bit slower so depending on what you're doing with a neural network a GPU actually isn't the right answer um and but the other cool thing about this which I think does have maybe even potential for there is that um you know one of the big not talked about problems in machine learning is that back propagation is the fastest thing we know but it's inherently not scalable like you can get a bigger better GPU to do your bits of your network in parallel to speed things up but that only gets better if you have a bigger network if you want to speed up the training process you can't just add another CPU or another GPU and make backdrop go faster you can make the forward and backward pass through your neural network faster but you still have to do but every Epoch of back prop iteratively um a method like this where we're generate we're not we're not one it's back prop three so it's not using back prop we can use a nature Inspire or other method to use hundreds of computers and you can throw twice as many computers at it and get a result twice as that twice as fast whereas that problem you can't do that so if you think about actually being able to train a neural network faster um that crop actually you know it's got a pretty pretty low speed limit for for what we need to do and it's kind of a big problem with the machine learning community that people don't like to talk about because they're like oh I'll just buy the next big Nvidia GPU and that'll do things faster so that that's super interesting does this maybe even bring up a kind of relationship where things like a graphics visualization of course a GPU does well and that's like the screen changing through time with the classical process that can be massively unfolded and then the cognitive models ultimately of the nest mates which which again can be nested but the cognitive the thing that's more Quantum more cognitive model like you can do in parallel because the minds are not influencing each other except through stigmergy so then that is CPU bound the size The Colony but um and then you could use different like Graphics techniques like there are colonies organisms so one ants being or or it's a philosophical question what what is the scale at which uh a exists but all throughout California with the Argentine Ant for example and so how do we deal with those kinds of like mesh work cognitive systems all the way on through 50 in an acorn there's just all these different trade-offs that are being made and like in the featureless deserts there's um different wayfinding pathfinding um sensor integration polarization of light like different cognitive strategies because they might be going out long distance and dragging something home not leaving any pheromone because it's not any more likely to have food there so in that case the stigma G is basically minimal to essentially none and then in other situations you could have something that's very adherent to distributions to the point of being like fit to to very um like kind of a normative path but that's happening at a level that allows the um different compute architectures different information architectures and ultimately different biological embodiments to to really engage fruitfully again looking at the variability the diversity of biological algorithms for Collective Behavior which have been studied by Professor Gordon and others in so many different angles yet sometimes it can feel like multi-agent models always start kind of like at Square One demonstrate some proof of concept phenomena and then that is utilized as part of like a bigger perspective but it's not like that model was ever like claimed to have been tuned to maximum performance it's like well we got decision making Behavior you could transfer this to group decision making with honeybee decision making or something like that but there's still a big gap there but I think what you're describing with the cans which is funny because it could be cannot but also the cans is the dialect which is spoken yeah so it was very I mean we came up with it great choice and it's just like yeah because there's multiple perspectives to swap from on the classical screen because the meaning of the word is is something that's happening that fourth dimension cognitively the meaning of the word isn't to be found just on the blanket just on the interface itself that's just the communication um and that's like a bounded systems and then if you model a cognitive system that doesn't have that kind of a constraint then so represented by a map that that has some kind of blanket uh index some kind of blanketing like if you don't embody that constraint in the statistical model the map you're ignoring one of the fundamental constraints of modeling the way that things happen in an embodied fashion maybe there's some abstract space for a certain problem that's just like a total slam dunk however for for full generality at least to the space that we know of of biological life forms and their engagements and like ecological engagements not just like within one Behavior just that space is is so vast and there's so much to learn across different systems within two again to abduce away into different like information architectures and active inference being some subsetter type of those so it's awesome work do you have any last comments um I uh well being giving ability to for friends to be self-aware and well aware about this environment it's actually something that we implemented um in in our last work here with the BP free accounts um they are indirectly aware about their environment and Theory um adapt adapting to the changes in their clinical environment by indicted meaning that they are evolving using a genetic based algorithm um to just change their behavior like how these things the the the word the word the pheromones when they take the steps and some other times or some other characters that they have so they are adapting but kind of like nuts in um an intelligent uh uh way but through it through through Evolution if you want to say actually we we consider putting a putting a brain in each one of these agents but then again as design because we couldn't like paralyze that it will take time to train each one of these brains as we evolve the neural awesome Alexander or Travis any last thoughts oh okay jeez I I I'm just really really happy I mean I think this this work is really interesting and it opens up a lot of um pretty cool Avenues again if we can get to the point where we're evolving colonies that are producing um and can see where that can go so one of the big issues in neural architecture search is the whole question of what is an optimal neural network and what an optimal neural network is um could be different for well will be different for different tasks but not only that even if it's the same data set how you're using that neural network could lead to less optimal lesser more optimal neural network right depending on doing with it maybe you need one that's more energy efficient maybe you don't care about energy efficient efficiency or performance and you'll take a slower neural network but you need more accuracy so being able to have algorithms which can automate this whole process for us and tune them to what we actually want to use the neural network for is really important and I think one just having ant colony optimization be able to optimize the network for a problem was great but two if we can make it such that the algorithm itself is self-optimizing it really can streamline this whole process where right now if you're doing machine learning it can be kind of miserable you like make a neural network architecture you try it out see how well it does oh nope that didn't do do so well let me tweak a couple knobs um automating that process my whole life as a computer scientist is about being lazy about being smart about it so whatever I can optimize so that I don't have to do it over and over again seems like a good good use of my time so I'll be smart about having to do as little as possible in the future then I guess uh my I don't have too much to add to that I think uh a lot of the good discussion has happened already and we talked about various implications and ways of viewing you know and Colony optimization from you know other perspectives including an active inference point of view so I guess really more from a closing thought on my end is that uh it will be interesting or it is an interesting direction to think about like I said earlier or suggested about the adaptation of the metaphor right to other systems and what are you trying to model and you know what's your goals from a scientific and philosophical point of view what are what are the questions you seek to answer and I think it might be very interesting again given other developments and you know Computing technology uh in ways in which you implement the parallelization that I think is That's What attracted me the most to a lot of these meta heuristic algorithms even things like particle swarm and you know when I worked with Travis many years ago on the exam algorithms you saw our names on that and uh working on that type of stuff the part that always caught my attention is again that ability to say I can put these entities on different you know processing Computing processing you know resources or devices and then they will interact or interact and exchange their results in some way to try to optimize uh some you know often complex multi-objective cost function and so I think the part that we'll see or that would encourage the wider spread adoption of even like meta-heuristic algorithms in general not to say that they aren't used a lot in for example the engineering domains is again the development of you know parallel Computing processing systems and I think exploiting things like asynchronous Computing that was again another angle that caught my attention from Travis you know he's done a lot of work on you know genetic optimization in their evolution from an asynchronous point of view and how can we allocate whatever resources are available you know and you know Distributing them across Global networks I think that might be the best shot to scaling up let's see with what we got right now there might be again you've mentioned annual Quantum technology Quantum Computing is another interesting place that sort of like changes the game um uh but you know barring you know technologies that we don't necessarily have exactly at their best at this moment you know how can we take advantage of Citizen citizen science or distributed computing or peer-to-peer type of communication and building massive active inference systems that you know embody like the multi-agent metaphor of and Colony optimization or other nature inspired Frameworks and um Can this system you know evolve over very long spans of time just like Evolution really worked another piece my final count and uh is that uh why I'm interested sometimes in evolution is that to me it is the inductive bias that provided us with structures that allow for example a human agent babies can you know to operate already babies can already recognize faces right and uh we we have certain instinctual reactions and certain mechanisms that evolutionists endowed us with and so a fascinating question is what is the interplay of this you know the idea of simulating an artificial form of evolution maybe building you know DNA structures are very very simplified computational structures which would answer Abdel rahman's concern about well maybe we don't want the agents to be too smart and of themselves because I can't really simulate that unless you give me like a decade to run the simulator um but you could maybe come up with a more fundamental primitive and then use that as a starting point for your neural network let's say Daniel you're you want to do some task in image segmentation and and you know like okay but what can uh your evolutionary framework give me well I'll say here here's a template to start from this is this is a kernel on which you build your framework and you know it's like a DNA structure and this has evolved across you know many many years of distributed peer-to-peer Computing and you can imagine building this Mammoth evolving continual learning style you know evolutionary algorithm whether it is based on uh genetic algorithms brand colony and you could imagine that might be an interesting way to think about it and by the way I am spitballing and generating an idea of like how I could Envision a scalable form without inventing a new Computing system that I don't know will or will not exist uh because Quantum has a lot of problems still to solve too like super conducting there's super temperatures or trapping photons as I have learned um so that might be an interesting Direction and I think the scaling of this especially from the Practical end is going to be the most important we're going to need to pull together all the tools that we have as I mentioned before so I'll stop there too because I'll end up rambling more so hopefully that made sense well this was very epic and inspiring so good luck with the work you're all welcome to suggest uh another piece that we might focus on or or continue the discussion however you see fit because it's super interesting Direction so thank you until next time thank you so much bye